CHAPTER 10

CHEMICAL COORDINATION AND INTEGRATION

CHAPTER ESSENCE AT A GLANCE:
Endocrine hormone feedback loops, membrane versus intracellular receptor cascades, and metabolic homeostasis.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Dual Mechanisms of Hormone Action:** Peptide and protein hormones (Insulin, Glucagon, Pituitary hormones, Epinephrine) are water-soluble, cannot traverse the lipid bilayer, and bind cell-surface membrane receptors to generate second messengers (cAMP, IP3, Ca2+) that amplify biochemical cascades. Lipid-soluble steroid hormones (Estrogen, Progesterone, Testosterone, Cortisol) and Iodothyronines cross plasma membranes, binding intracellular nuclear receptors to modulate gene transcription directly.

• **Hypothalamus-Pituitary Endocrine Axis:** Hypothalamus controls pituitary function via releasing hormones (GnRH) and inhibiting hormones (Somatostatin). Anterior pituitary (Adenohypophysis) secretes GH, TSH, ACTH, PRL, LH, and FSH; Posterior pituitary (Neurohypophysis) stores and releases Oxytocin (triggers uterine contraction during parturition and milk ejection) and Vasopressin (ADH) (triggers water conservation in kidney collecting ducts).

• **Calcium and Glucose Homeostatic Antagonisms:** Calcium homeostasis: Parathyroid hormone (PTH) is hypercalcemic (increases blood Ca2+ by promoting bone resorption and renal or gut reabsorption), acting antagonistically to Thyrocalcitonin (TCT), which is hypocalcemic. Glucose homeostasis: Pancreatic islets secrete Glucagon from alpha cells (hyperglycemic, stimulates glycogenolysis and gluconeogenesis) and Insulin from beta cells (hypoglycemic, stimulates cellular glucose uptake and glycogenesis).

• **Adrenal Gland and Endocrine Pathologies:** Adrenal medulla secretes emergency fight-or-flight catecholamines (Epinephrine, Norepinephrine) that increase heart rate, alertness, and glycogenolysis. Adrenal cortex secretes Glucocorticoids (cortisol) and Mineralocorticoids (aldosterone). Endocrine disorders: Acromegaly, Gigantism, or Dwarfism (GH), Cretinism or Graves's disease (Thyroid), Addison's disease (Adrenal cortex hyposecretion), and Diabetes mellitus (Insulin deficiency causing hyperglycemia, glycosuria, and ketonuria).

ESSENTIAL PHYSIOLOGICAL PATHWAYS & STRUCTURES

$$\text{Signaling Modality} = \frac{\text{Peptide: Cell Surface Receptor} \rightarrow \text{Second Messenger (cAMP)}}{\text{Steroid: Intracellular Receptor} \rightarrow \text{Gene Transcription}}$$

Peptide vs Steroid Hormone Molecular Action

$$\text{Calcium Axis} = \text{PTH (Parathyroid: Hypercalcemic)} \overset{\text{Antagonistic}}{\longleftrightarrow} \text{TCT (Thyroid: Hypocalcemic)}$$

Antagonistic Endocrine Control of Serum Calcium

$$\text{Glucose Axis} = \text{Insulin (Beta-Cells: Hypoglycemic)} \overset{\text{Antagonistic}}{\longleftrightarrow} \text{Glucagon (Alpha-Cells: Hyperglycemic)}$$

Homeostatic Glucose Regulation in Pancreatic Islets

$$\text{Fight-or-Flight} = \text{Epinephrine} \rightarrow \text{Heart Rate}\uparrow + \text{Pupil Dilation} + \text{Glycogenolysis}\uparrow$$

Sympathomimetic Responses of the Adrenal Medulla

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Insulin decreases blood glucose (hypoglycemic), while Glucagon increases blood glucose (hyperglycemic); PTH increases blood calcium, while Calcitonin decreases blood calcium.

• **Alert:** Diabetes mellitus is caused by insulin deficiency leading to high blood glucose and ketone bodies; Diabetes insipidus is caused by ADH deficiency leading to dilute polyuria without hyperglycemia.

• **Alert:** Oxytocin and Vasopressin are synthesized by the hypothalamus and merely stored and released by the posterior pituitary (neurohypophysis).

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Distinguish between the mechanisms of action of a peptide hormone (e.g., Epinephrine) and a steroid hormone (e.g., Estrogen) regarding receptor location and second messengers. [Score: 2 Marks]

Answer & Key Valuation Steps:

• Peptide hormone: Binds to extracellular cell-surface membrane receptors; does not enter the target cell, generating intracellular second messengers (cAMP, IP3, Ca2+) to regulate cellular metabolism. **[1 Mark]**

• Steroid hormone: Diffuses across the lipid membrane, binds to intracellular nuclear receptors, and the hormone-receptor complex interacts directly with the genome to regulate gene transcription. **[1 Mark]**

2. Explain how Insulin and Glucagon act antagonistically to maintain blood glucose homeostasis. [Score: 2 Marks]

Answer & Key Valuation Steps:

• Insulin (Beta-cells): Hypoglycemic hormone; acts on hepatocytes and adipocytes to enhance cellular glucose uptake and utilization, stimulating glycogenesis (conversion of glucose to glycogen). **[1 Mark]**

• Glucagon (Alpha-cells): Hyperglycemic hormone; acts on liver hepatocytes to stimulate glycogenolysis (breakdown of glycogen to glucose) and gluconeogenesis, releasing glucose into the blood. **[1 Mark]**

3. Describe how Parathyroid Hormone (PTH) and Thyrocalcitonin (TCT) regulate calcium levels in the human body. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Parathyroid Hormone (PTH): Hypercalcemic hormone; increases blood Ca2+ levels by stimulating bone resorption (osteoclasts dissolving bone matrix) and promoting reabsorption of Ca2+ by renal tubules and digested food in the intestine. **[1.5 Marks]**

• Thyrocalcitonin (TCT): Hypocalcemic hormone secreted by thyroid C-cells; decreases blood Ca2+ levels by suppressing bone resorption and promoting calcium deposition into bone matrix, acting antagonistically to PTH. **[1.5 Marks]**

4. Outline the physiological functions of Oxytocin and Vasopressin synthesized by the hypothalamus and released by the neurohypophysis. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Oxytocin: Acts on smooth muscles of the uterus, stimulating vigorous contractions during parturition (childbirth); also stimulates milk ejection from mammary glands during suckling. **[1.5 Marks]**

• Vasopressin (ADH): Acts mainly at the kidneys, stimulating active water reabsorption from distal tubules and collecting ducts, preventing water loss and diuresis. **[1.5 Marks]**

5. Explain why the hormones of the adrenal medulla (Epinephrine and Norepinephrine) are called the emergency hormones or fight-or-flight hormones. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Secreted rapidly in response to stress of any kind and during emergency fight-or-flight situations. **[1 Mark]**

• Increase alertness, pupillary dilation, piloerection (raising of hairs), and sweating. **[1 Mark]**

• Increase heartbeat, strength of heart contraction, and rate of respiration; stimulate breakdown of glycogen, lipids, and proteins to raise blood glucose concentrations for muscle readiness. **[1 Mark]**

6. Identify the endocrine hormone responsible for each condition: (a) Acromegaly in adults, (b) Cretinism in infants, (c) Exophthalmic goitre (Grave's disease), (d) Addison's disease. [Score: 3 Marks]

Answer & Key Valuation Steps:

• (a) Acromegaly: Hypersecretion of Growth Hormone (GH) in adults. **[0.75 Mark]**

• (b) Cretinism: Hyposecretion of Thyroid hormones (thyroxine) during pregnancy and infancy. **[0.75 Mark]**

• (c) Exophthalmic goitre (Grave's disease): Hypersecretion of Thyroid hormones (hyperthyroidism). **[0.75 Mark]**

• (d) Addison's disease: Hyposecretion of adrenal cortex hormones (mineralocorticoids and glucocorticoids). **[0.75 Mark]**

7. A 48-year-old patient presents with excessive thirst (polydipsia), frequent urination (polyuria), high fasting blood glucose, and the presence of ketone bodies in urine. Identify the disorder, specify the deficient hormone, identify the producing cells, and explain the metabolic origin of ketonuria. [Score: 4 Marks]

Answer & Key Valuation Steps:

• Disorder identification: Diabetes Mellitus (distinguished from Diabetes Insipidus by hyperglycemia and glycosuria). **[1 Mark]**

• Deficient hormone and source: Insulin, secreted by the beta cells of the Islets of Langerhans in the pancreas. **[1 Mark]**

• Pathophysiology: Lack of insulin prevents target tissues from taking up and utilizing circulating glucose, causing glucose accumulation in blood (hyperglycemia) exceeding the renal threshold, resulting in glycosuria. **[1 Mark]**

• Metabolic origin of ketonuria: Deprived of intracellular glucose, cells resort to accelerated breakdown of fats (lipolysis) for energy; excess fatty acid oxidation in the liver produces acetoacetate and beta-hydroxybutyrate (ketone bodies), excreted in urine as ketonuria. **[1 Mark]**